

Chapter - 4

Types of Data in Cluster Analysis

- **1. Interval-Scaled Data**
- **Definition:** Continuous numerical data measured on an interval scale.
- **Examples:** Age, income, temperature.
- **Distance measure:** Euclidean or Manhattan distance.
- **2. Binary Data**
- **Definition:** Variables with only two possible values — 0 or 1 (yes/no, true/false).
- **Types:**
 - **Symmetric:** Both outcomes are equally important (e.g., gender).
 - **Asymmetric:** Only one outcome (1) is important (e.g., presence of a symptom).

3. Categorical (Nominal) Data

- **Definition:** Data that represent categories with no inherent order.
- **Examples:** Eye color, type of car, city.
- **Similarity measure:** Simple matching coefficient.

4. Ordinal Data

- **Definition:** Categorical data with a natural order or ranking.
- **Examples:** Satisfaction level (low, medium, high), education level.
- **Approach:** Convert ranks into numeric scores before clustering.

5. Ratio-Scaled Data

- **Definition:** Numeric data with a true zero point.
- **Examples:** Height, weight, distance, salary.
- **Approach:** Often normalized before clustering.

6 Mixed-Type Data

- **Definition:** Combination of different data types (numerical, categorical, binary).
- **Example:** A dataset with age (numeric), gender (binary), and city (categorical).
- **Approach:** Use special distance measures like **Gower's distance**.
- Would you like me to add a **short summary table** for quick revision?

What Are Partitioning Methods?

- Partitioning methods are a type of **clustering technique** used in **data mining and machine learning**.
- They **divide (partition)** a dataset into a fixed number of groups (called **clusters**) so that:
- **Data points in the same cluster** are **similar** to each other.
- **Data points in different clusters** are **different** from each other.

K-Means Clustering

- K-Means divides the data into **k clusters** based on the **mean (average)** value of the points in each cluster.

Steps :

1. **Choose k** — the number of clusters you want.
2. **Pick k points randomly** from your data. These are called **centroids** (cluster centers).
3. **Assign each data point** to the nearest centroid (based on distance, usually Euclidean distance).
4. **Recalculate the centroid** of each cluster (take the average of all points in that cluster).
5. **Repeat steps 3 and 4** until the centroids stop changing much (or after a few iterations).

- Suppose you want to group students based on **marks in Maths and Science**:
- Choose $k = 3$ (you want 3 clusters).
- The algorithm groups students into 3 clusters:
 - **High performers**
 - **Average performers**
 - **Low performers**
- Each group's **centroid** represents the average performance of that group.

Advantages:

- Simple and fast.
- Works well for large datasets.

Disadvantages:

- Sensitive to **outliers** (one extreme value can shift the mean).
- You must **choose k** in advance.
- Works best with **numerical data**.

K-Medoids Clustering (PAM: Partitioning Around Medoids)

- K-Medoids is **similar to K-Means**, but instead of using the **mean**, it uses a **medoid** — a real data point that is **most central** to the cluster.
- So, the **medoid** is an **actual data point** (not an average).

Steps:

1. **Choose k** (number of clusters).
2. **Select k actual data points** as medoids.
3. **Assign each data point** to the nearest medoid.
4. **Try swapping** a medoid with a non-medoid data point.
5. **Check the total cost (distance)** — if it improves, keep the new medoid.
6. Repeat until the clusters are stable.

Example:

- Using the same student example:
- Choose 3 medoids (actual students representing each group).
- Other students are assigned to the medoid they are most similar to.

Advantages:

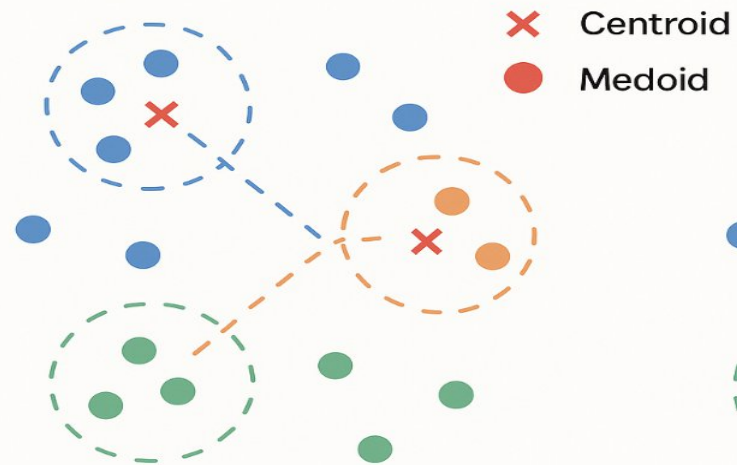
- **Less sensitive to outliers** (since it uses actual data points, not averages).
- Works with **categorical data** (not just numbers).

Disadvantages:

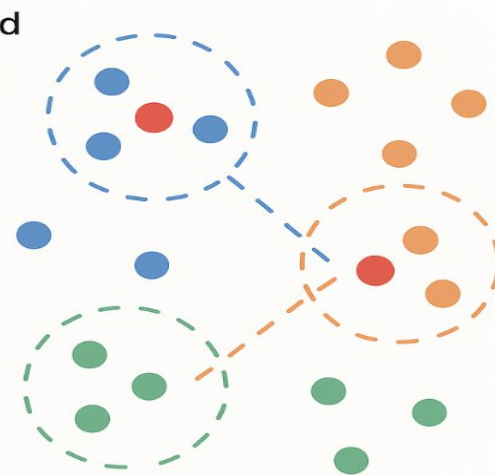
- **Slower** than K-Means for large datasets.
- More complex to compute.

Partitioning Methods

k-Means



k-Medoids



Left Side – K-Means

- The **red X** marks are **centroids** — the *average position* of all points in that cluster.
- Each **color group (blue, orange, green)** shows points that are *close to that centroid*.
- Centroids are **not actual data points**, just calculated averages.

Right Side – K-Medoids

- The **red circles** are **medoids** — *real data points* that are most central to their cluster.
- The clusters (blue, orange, green) are grouped around these medoids.
- Since medoids are real points, they are **less affected by outliers**.
- **K-Means** → uses *average points (centroids)*
- **K-Medoids** → uses *actual representative points (medoids)*

Example- We have 5 students and their marks in **Maths and Science**.

Student	Maths	Science
A	85	80
B	82	78
C	25	30
D	28	28
E	70	65

- We want to divide them into **2 groups ($k = 2$)** — for example:
- High-scoring students
- Low-scoring students
- **Step 1: Choose 2 starting points (centroids)**
- Let's pick:
- **Centroid 1** = A (85, 80) → high scores
- **Centroid 2** = C (25, 30) → low scores
- These are our initial “cluster centers.”

⚙️ Step 2: Find which centroid each student is closest to

We use **Euclidean distance** formula:

$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Let's calculate for each student 📌

◆ Student A (85, 80)

- Distance from Centroid 1 (85,80):

$$\sqrt{(85 - 85)^2 + (80 - 80)^2} = 0$$

- Distance from Centroid 2 (25,30):

$$\sqrt{(85 - 25)^2 + (80 - 30)^2} = \sqrt{3600 + 2500} = \sqrt{6100} \approx 78.10$$

✅ Closer to Centroid 1 → belongs to **Cluster 1**

◆ **Student B (82, 78)**

- Distance from (85,80):

$$\sqrt{(82 - 85)^2 + (78 - 80)^2} = \sqrt{9 + 4} = \sqrt{13} \approx 3.6$$

- Distance from (25,30):

$$\sqrt{(82 - 25)^2 + (78 - 30)^2} = \sqrt{3249 + 2304} = \sqrt{5553} \approx 74.5$$

✓ Closer to Centroid 1 → **Cluster 1**

◆ **Student C (25, 30)**

- Distance from (85,80):

$$\sqrt{(25 - 85)^2 + (30 - 80)^2} = 78.10$$

- Distance from (25,30):

$$\sqrt{(25 - 25)^2 + (30 - 30)^2} = 0$$

✓ Closer to Centroid 2 → **Cluster 2**



◆ **Student D (28, 28)**

- Distance from (85,80):

$$\sqrt{(28 - 85)^2 + (28 - 80)^2} = \sqrt{3249 + 2704} = \sqrt{5953} \approx 77.2$$

- Distance from (25,30):

$$\sqrt{(28 - 25)^2 + (28 - 30)^2} = \sqrt{9 + 4} = \sqrt{13} \approx 3.6$$

✓ Closer to Centroid 2 → **Cluster 2**

◆ **Student E (70, 65)**

- Distance from (85,80):

$$\sqrt{(70 - 85)^2 + (65 - 80)^2} = \sqrt{225 + 225} = \sqrt{450} \approx 21.2$$

- Distance from (25,30):

$$\sqrt{(70 - 25)^2 + (65 - 30)^2} = \sqrt{2025 + 1225} = \sqrt{3250} \approx 57.0$$

✓ Closer to Centroid 1 → **Cluster 1**

Step 3: Form clusters

Cluster 1

A (85,80)

B (82,78)

E (70,65)

Cluster 2

C (25,30)

D (28,28)

Step 4: Calculate new centroids (average)

For each cluster, find the **average Maths and Science marks**.

Cluster 1 (A,B,E):

$$\text{New Centroid 1} = \left(\frac{85 + 82 + 70}{3}, \frac{80 + 78 + 65}{3} \right) = (79, 74.33)$$

Cluster 2 (C,D):

$$\text{New Centroid 2} = \left(\frac{25 + 28}{2}, \frac{30 + 28}{2} \right) = (26.5, 29)$$

Step 5: Recalculate distances (optional)

You can repeat the distance calculation using the **new centroids** to see if any student moves to another cluster.

Here, all remain in their groups → clustering is complete 

Cluster

Students

Meaning

Cluster 1

A, B, E

High scorers

Cluster 2

C, D

Low scorers

- **K-Medoids Clustering**
- Instead of using *mean*, we use *actual data points (medoids)* as centers.
- **Step 1: Choose initial medoids**
- Let's pick:
- Medoid 1 = A (85,80)
- Medoid 2 = C (25,30)
- **Step 2: Assign students to nearest medoid**
- (Same result as before)
- **Cluster 1:** A, B, E
Cluster 2: C, D
- **Step 3: Check if swapping medoids reduces total distance**
- We try replacing medoid 1 or 2 with another data point in the same cluster to see if the overall distance decreases.
- In this case, A and C remain the best medoids (lowest total distance).
- Final Clusters:
- Cluster 1 (Medoid = A): A, B, E
- Cluster 2 (Medoid = C): C, D

Feature	K-Means	K-Medoids
Cluster center	Mean (average) of points	Medoid (actual data point)
Suitable for	Numerical data	Numerical or categorical data
Sensitive to outliers	Yes	No
Speed	Fast	Slower
Accuracy	Less accurate with noise	More accurate with noise

Example: K-Medoids Clustering (2 clusters)

Student	Maths	Science
A	85	80
B	82	78
C	25	30
D	28	28
E	70	65

- We want to divide them into **2 clusters ($k = 2$)**.
- **Step 1: Choose initial medoids**
- In **K-Medoids**, we don't use averages — we pick **actual data points** as the centers (called **medoids**).
- Let's start with:
- **Medoid 1** = A (85, 80)
- **Medoid 2** = C (25, 30)

- **Step 2: Assign each student to the nearest medoid**
- We'll again use **Euclidean distance**, just like in K-Means, but our "centers" are real points (A and C).

Student	Distance from A (85,80)	Distance from C (25,30)	Nearest Medoid	Cluster
A	0	78.10	A	1
B	3.6	74.5	A	1
C	78.1	0	C	2
D	77.2	3.6	C	2
E	21.2	57.0	A	1

Cluster 1 (Medoid A): A, B, E

Cluster 2 (Medoid C): C, D

- **Check the total cost (distance)**
- Let's find the **total distance** of all students from their medoid.
- **For Cluster 1 (Medoid = A):**

Student	Distance to A
A	0
B	3.6
E	21.2
Total = 0 + 3.6 + 21.2 = 24.8	

- For Cluster 2 (Medoid = C):

Student

Distance to C

C

0

D

3.6

Total = 3.6

Overall cost = $24.8 + 3.6 = 28.4$

- **Step 4: Try swapping medoids**
- Now we check if choosing a **different medoid** (like B or E instead of A) **reduces the total cost**.
- Let's test one swap
- Suppose we make **B (82,78)** the medoid instead of A.

New distances for Cluster 1 (Medoid = B):

Student	Distance to B
A	3.6
B	0
E	18.0
Total = 21.6 (less than 24.8 earlier!)	

So B is a better medoid than A

- **Step 5: Final clusters**
- After testing possible swaps, we end up with:
- **Cluster 1 (Medoid = B): A, B, E**
- **Cluster 2 (Medoid = C): C, D**
- This is our **final K-Medoids result**.

Feature

Center type

Sensitive to outliers

Speed

Used for

Example result

K-Means

Average (not real)

Yes

Fast

Numerical data

Centroids at (79,74) & (26.5,29)

K-Medoids

Actual data point

No

Slower

Numerical or categorical data

Medoids at **B (82,78)** & **C (25,30)**

Method

Cluster 1

Cluster 2

K-Means

A, B, E

C, D

K-Medoids

A, B, E

C, D